Select under one law; evaluate under another

Hold the problem fixed

y, targets, loss, preferences;
M; nonempty finite family F(y).

1. Choose under supplied beliefs

$\widehat{g} \in \mathcal{F}(y)$
$R^s(\widehat{g}) \leq \min_{g \in \mathcal{F}} R^s(g) + \eta$
Actual selection-risk gap η ≥ 0.

2. Freeze the selected report

Same complete report ĝ in both risks.

3. Assess the same report twice

Supplied law $\Pi_s(\cdot \mid y)$: $R^s(\widehat{g})$

Evaluation law $P(\cdot \mid y)$: $R^P(\widehat{g})$

P is a declared evaluation model;
its correctness is not established here.

A conditional common-family bound

$d = \max_{g \in \mathcal{F}} |R^s(g) - R^P(g)|$
$R^P(\widehat{g}) - \min_{g \in \mathcal{F}} R^P(g) \leq 2d + \eta$
The diagram supplies no value for d.

Numerical approximation

Approximate the same supplied law.
More draws can improve integration;
they do not establish agreement with P.

Discrepancy between laws

Requires evidence about the model.
Different laws can share action risks.
Risk control does not transfer
criterion protections across laws.

Neutral loss also retains distinctness
and withholding; guard families must match.